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Visualise the embeddings: simplebiosnap.streamlit.app
Paper preprint: arxiv.org/abs/2404.11374

# Efficient Polypharmacy Side Effect Prediction Using Tensor Factorisation

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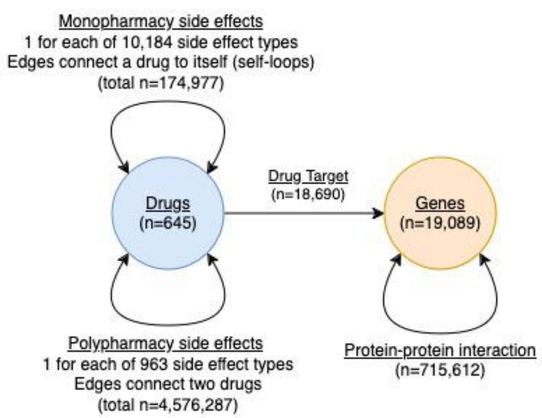
## Background

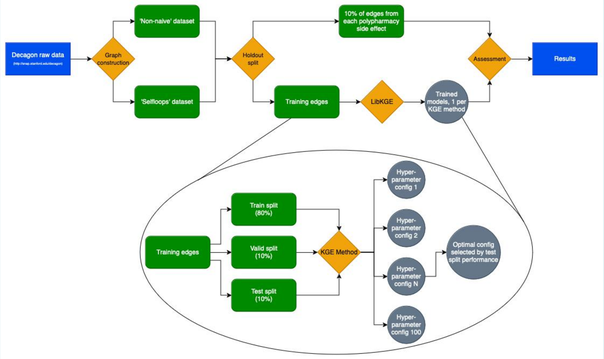
- **Prediction of adverse drug interactions** is an increasingly important task due to ageing populations across the globe
- **Computational methods** are essential due to the combinatorial nature of the problem
- However, state-of-the-art research in this area is dominated by techniques that require **large amounts of processing power** and **ever-growing datasets** – a phenomenon referred to as "Red AI" (Schwartz *et al*, 2020)
- This means that polypharmacy side effect prediction research has a high carbon footprint, a high barrier to entry for researchers trying to reproduce with fewer resources, and **reduced real-world applicability to poorer communities**

## Data

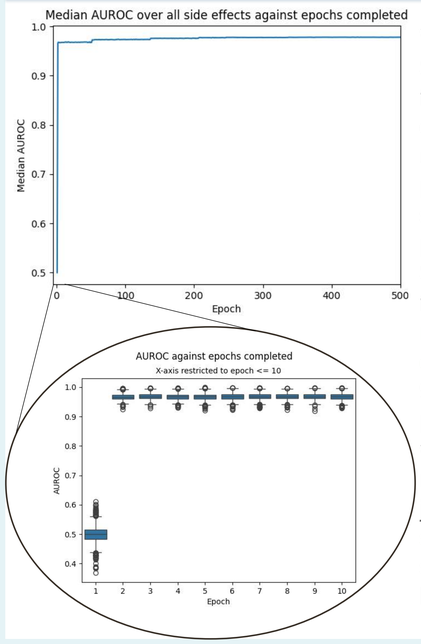



The data is the same used by Zitnik *et al.* (2018) but we include the monopharmacy data as edges in the graph rather than initialising vectors with it.

## Experiment Workflow

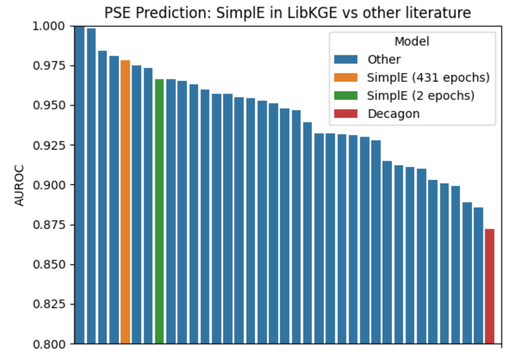



## Training Progression



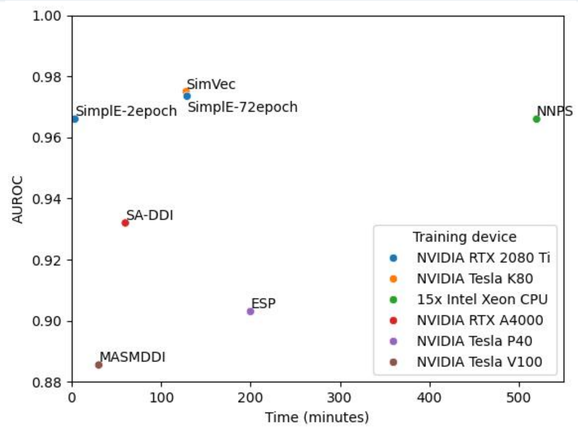

We allowed SimplE to train for 500 epochs in order to see its upper limit for polypharmacy side effect prediction. Suprisingly, we found that it only needed two epochs to reach 99% of the performance it would ever reach. This took just 214.7 seconds on a single NVIDIA RTX 2080 Ti.

## Maximum Performance




After 431 epochs, performance on holdout data maxed out at 0.978 AUROC – this puts SimplE **fifth place in the literature**.

## Performance vs. Training Time




Among papers that report training time, SimplE after two epochs outperforms all but one. SimVec scored 0.009 AUROC higher than that model, but needed more than 35x as much training time to do so.

## Discussion and Takeaways

- SimplE represents an opportunity for a substantial step towards "Green AI" in the field of polypharmacy side effect prediction
- When trained on a six year-old GPU, SimplE produces highly predictive embeddings in less than four minutes. No other model in the literature reports a performance:cost ratio close to this
- This result is also achieved with less data than comparable methods as the input graph is in RDF format
- Red AI has produced many valuable insights and should continue to have a prominent place in the field. Microsoft's DSN-DDI is an impressive example of the upper predictive limit of complex models such as graph neural networks
- Despite such successes, we call for more transparency in the field regarding the computational burden of presented models, and higher representation of models with linear complexity, such as tensor factorisation approaches

## Acknowledgements

Decagon: Zitnik, Agrawal, and Leskovec, 2018

Libkge: Broscheit *et al*, 2020

SimplE: Kazemi and Poole, 2018

Green AI: Schwartz *et al*, 2020

SimVec: Lukashina *et al*, 2021

DSN-DDI: Li *et al*, 2023

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